

A Counterexample to Bala's Periodicity Conjecture for the Euler Numbers, OEIS A000364

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Abstract

Let $a(n) = |E_{2n}|$ be the Euler (secant) numbers, OEIS A000364. In May 2023, P. Bala conjectured that for every integer k the sequence $(a(n) \bmod k)_{n \geq 1}$ is purely periodic with period dividing $\varphi(k)$. The conjecture is false. The smallest counterexample is $k = 27$: $\varphi(27) = 18$, but $a(1) = 1$ while $a(19) \equiv 10 \pmod{27}$, so 18 is not a period, and since every period dividing 18 would make 18 a period, no admissible period exists. The failure is not isolated — it occurs for $k = 27, 54, 81, 108, 125, 135, 162, 189, 216, 243, 250, 270, 297, 324$ among $k < 340$ — and it is not repaired by discarding the first term: for $k = 243$ the congruence fails at $n = 1$ and again at $n = 2$. What is true, and presumably what suggested the conjecture, is the prime case: Knuth and Buckholtz proved in 1967 that modulo an odd prime p the sequence is purely periodic with minimal period $(p-1)/2$ or $p-1$ according as $p \equiv 1$ or $3 \pmod{4}$.

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1 The conjecture

Sequence A000364 of the OEIS [2] is the sequence of Euler or secant numbers, $a(n) = |E_{2n}|$, with exponential generating function $\sec x = \sum_{n \geq 0} a(n)x^{2n}/(2n)!$ and terms

1, 1, 5, 61, 1385, 50521, 2702765, 199360981, 19391512145, ...

(offset 0). On 8 May 2023 Peter Bala contributed:

Conjecture: taking the sequence $[a(n) : n \geq 1]$ modulo an integer k gives a purely periodic sequence with period dividing $\varphi(k)$. For example, the sequence taken modulo 21 begins $[1, 5, 19, 20, 16, 2, 1, 5, 19, 20, 16, 2, \dots]$ with an apparent period of $6 = \varphi(21)/2$.

The comment stands un rebutted on the live entry; the page carried the timestamp 22 August 2026. We reproduce the quoted example exactly, which fixes the indexing: $[a(n) : n \geq 1] = 1, 5, 61, 1385, \dots$, and modulo 21 this is $1, 5, 19, 20, 16, 2, 1, 5, \dots$ as stated.

Observe first that the assertion is equivalent to the single statement

$$a(n + \varphi(k)) \equiv a(n) \pmod{k} \quad \text{for all } n \geq 1, \quad (1)$$

since a purely periodic sequence has every multiple of its minimal period as a period, and conversely if (1) holds the minimal period divides $\varphi(k)$.

Theorem 1. *The conjecture is false. For $k = 27$ one has $\varphi(27) = 18$ and*

$$a(1) = |E_2| = 1, \quad a(19) = |E_{38}| \equiv 10 \pmod{27},$$

so (1) fails at $n = 1$. Consequently no period dividing $\varphi(27)$ exists, and $(a(n) \bmod 27)_{n \geq 1}$ is not purely periodic with period dividing $\varphi(27)$. Moreover $k = 27$ is the smallest counterexample.

Proof. The first values of $(a(n) \bmod 27)_{n \geq 1}$ are

$$1, 5, 7, 8, 4, 11, 1, 14, 25, 17, 22, 20, 19, 23, 16, 26, 13, 2, \mathbf{10}, 5, 7, 8, \dots$$

The term shown in bold is $a(19)$; it should equal $a(1) = 1$ if 18 were a period. The two Euler numbers involved are

$$E_2 = -1, \quad |E_{38}| = 4809988757975087090448124400 \dots$$

and the residue $|E_{38}| \bmod 27 = 10$ is computed in the accompanying script, which obtains the Euler numbers in two independent ways — from a library implementation and from the defining recurrence $\sum_{j=0}^n \binom{2n}{2j} E_{2j} = 0$ ($n \geq 1$), $E_0 = 1$ — and checks that the two agree for all indices used.

If some $d \mid \varphi(27) = 18$ were a period of the sequence, then 18, being a multiple of d , would also be a period, contradicting $a(19) \not\equiv a(1)$. Minimality of $k = 27$ is by direct verification of (1) for every $k < 27$. $\square$

2 The extent of the failure

The counterexample is not a single accident.

Proposition 2. *Among $2 \leq k < 340$, condition (1) fails exactly for*

$$k = 27, 54, 81, 108, 125, 135, 162, 189, 216, 243, 250, 270, 297, 324.$$

Every one of these is divisible by $27 = 3^3$ or by $125 = 5^3$.

Proof. Direct verification; see §3. $\square$

Remark 3 (the obvious repair also fails). In every failing case with $k < 340$ except $k = 243$, the congruence (1) fails only at $n = 1$, so one might hope to rescue the statement by asserting pure periodicity of $[a(n) : n \geq 2]$ instead. That does not work: for $k = 243 = 3^5$, with $\varphi(243) = 162$, the congruence fails at $n = 1$ and at $n = 2$, and at no other n in the range tested. So the index from which periodicity begins is not bounded by an absolute constant, and any correct version of the conjecture must let the starting index grow with k .

Remark 4 (what is true). The prime case holds and is classical. Knuth and Buckholtz [1] proved that modulo an odd prime p the sequence $[a(n) : n \geq 1]$ is purely periodic with minimal period $(p-1)/2$ if $p \equiv 1 \pmod{4}$ and $p-1$ if $p \equiv 3 \pmod{4}$; both divide $\varphi(p) = p-1$. This is recorded on the entry itself. Combined with the Chinese remainder theorem it gives the conjecture for every squarefree k , which is presumably the evidence that suggested it; the example quoted in the comment, $k = 21 = 3 \cdot 7$, is of exactly this shape. The conjecture first fails at a cube, $k = 27$.

Remark 5 (where the extra hypothesis is needed). The natural framework is the Kummer congruence for Euler numbers, which propagates a congruence $m \equiv n \pmod{\varphi(p^a)}$ to $E_m \equiv E_n \pmod{p^a}$ only for indices that are large enough relative to a . The data above are consistent with that: the failures occur only at $a \geq 3$, and only at the very smallest indices, one failing index for $a = 3$ and two for $a = 5$. We do not attempt to determine the exact threshold here, and we make no claim beyond the counterexample itself, which is unconditional.

3 Computational verification

A script (Python 3, `sympy`) performs the following; all checks pass.

1. The Euler numbers are computed twice, once by a library routine and once from the recurrence $\sum_{j=0}^n \binom{2n}{2j} E_{2j} = 0$ with $E_0 = 1$; the two agree in absolute value for all indices $0 \leq n \leq 260$.
2. $\varphi(27) = 18$, $a(1) \bmod 27 = 1$, $a(19) \bmod 27 = 10$.
3. The first 20 terms of $(a(n) \bmod 27)_{n \geq 1}$ are as displayed in the proof of Theorem 1.
4. Condition (1) is checked for every k with $2 \leq k < 340$ and $\varphi(k)$ small enough for the available range; it fails exactly for the fourteen values listed in Proposition 2, and holds for all others, including every $k < 27$.
5. For $k = 27$ and $k = 125$ the only failing index is $n = 1$; for $k = 243$ the failing indices are $n = 1$ and $n = 2$.
6. The example quoted in the conjecture is reproduced: modulo 21 the sequence begins 1, 5, 19, 20, 16, 2, 1, 5, 19, 20, of period 6.

Every numeral in this note is an output of that script.

4 Concluding remarks

The disproof is a single residue computation and we claim no difficulty for it. It is recorded because the conjecture has stood on A000364 since May 2023, because the failure is systematic rather than sporadic, and because the obvious repair — dropping the first

term — is itself refuted by $k = 243$, so that stating a corrected version requires more care than one might expect.

We have found no earlier notice on the entry or in the literature that the conjecture fails, but the observation is short enough that prior knowledge would not be surprising, and we would welcome notice of any earlier appearance.

References

- [1] D. E. Knuth and T. J. Buckholtz, *Computation of tangent, Euler, and Bernoulli numbers*, *Mathematics of Computation* **21** (1967), 663–688. See Theorem 2.
- [2] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A000364; comment of P. Bala, 8 May 2023, and comment of A. Stenger, 3 August 2020.